

Probability & Statistics Complete Notebook

A complete study and revision reference for the Probability and Statistics step of the AI Research Roadmap — the last step of the maths stage.

What is inside

5 topics	Describing data → Bayes → distributions
Full lessons	Plain-language explanation of every concept
Diagrams	Bayes tree and the normal distribution
Worked examples	Every calculation written out step by step
Tests + answers	3 questions per topic, easy · normal · hard
Master reference	All key points in one table

Inference statistics — hypothesis testing, confidence intervals, p-values — is skipped at this stage. It is not required before Machine Learning.

Phase 1 · Stage 1 · Step 4

Describing data

Centre, spread, and why every ML pipeline starts here.

THE IDEA

Before you can model data you have to describe it. Two questions cover most of it: **where is the centre**, and **how spread out is it**?

Measure	What it tells you
Mean	The average. Add everything, divide by how many. Pulled by extreme values
Median	The middle value once sorted. Ignores extremes — better for skewed data like income
Variance	The average <i>squared</i> distance from the mean. Big variance = widely scattered data
Standard deviation	The square root of the variance — back in the original units

WHY VARIANCE SQUARES THE DISTANCES

If you just averaged the raw distances from the mean, the positives and negatives would cancel and you would always get zero. Squaring removes the signs. The square root at the end undoes the squaring so the answer is readable again — if your data is in years, the standard deviation is in years, but the variance is in years².

WORKED EXAMPLE

TOPIC 1 — DESCRIBING DATA

centre, spread, and why ML cares

OUR DATA — five patient ages

2, 4, 4, 6, 9

MEAN (the average)

add them up, divide by how many

$$2 + 4 + 4 + 6 + 9 = 25$$

$$25 / 5 = 5$$

MEDIAN (the middle value)

sort them, take the middle one : 2 4 [4] 6 9

$$\text{median} = 4$$

median ignores extreme values — mean does not

VARIANCE (how spread out)

step 1 : distance of each value from the mean (5)

$$2-5 = -3 \quad 4-5 = -1 \quad 4-5 = -1$$

$$6-5 = 1 \quad 9-5 = 4$$

step 2 : square each one (removes the minus signs)

$$9 + 1 + 1 + 1 + 16 = 28$$

step 3 : divide by how many

$$\text{variance} = 28 / 5 = 5.6$$

STANDARD DEVIATION

just the square root of the variance :

$$\sqrt{5.6} \approx 2.37$$

back in the original units — that is why we use it

ML LINK : normalising data is $(x - \text{mean}) / \text{std}$

it makes features with different scales comparable,
and it is the first step of almost every ML pipeline.

Why it matters for ML: normalisation is $(x - \text{mean}) / \text{std}$. Every feature you feed a model gets this treatment so that a feature measured in thousands does not drown one measured in decimals. Mean and standard deviation are also what "batch normalisation" normalises.

KEY POINTS FROM THIS TOPIC

Mean	add all values, divide by how many
Median	the middle value once sorted
Variance	average of the squared distances from the mean
Standard deviation	$\sqrt{\text{variance}}$ — back in original units
Normalisation	$(x - \text{mean}) / \text{std}$

TEST — Topic 1

Dataset: 3, 5, 7, 9, 11

1. Find the mean and the median. → **mean = 7, median = 7**
2. Find the variance. → **distances -4,-2,0,2,4 → squares 16,4,0,4,16 = 40 → 40/5 = 8**
3. Find the standard deviation, then normalise the value 9. → **std = $\sqrt{8} \approx 2.83$; $(9 - 7)/2.83 \approx 0.71$**

Probability basics

Measuring how likely something is.

THE IDEA

Probability is a number between **0** (impossible) and **1** (certain) that measures how likely an event is.

$$P(\text{event}) = \text{favourable outcomes} / \text{total outcomes}$$

THE THREE RULES YOU NEED

Rule	Formula	When
AND	$P(A \text{ and } B) = P(A) \times P(B)$	Independent events — one does not affect the other
OR	$P(A \text{ or } B) = P(A) + P(B)$	Events that cannot both happen
OR (overlapping)	$P(A) + P(B) - P(A \text{ and } B)$	Events that can both happen — subtract the double count
Complement	$P(\text{not } A) = 1 - P(A)$	When "at least one" is easier counted backwards

The complement rule is the most useful shortcut in the whole topic. "At least one head in two coins" would need three cases counted directly; backwards it is one calculation: $1 - P(\text{no heads}) = 1 - \frac{1}{4} = \frac{3}{4}$.

WORKED EXAMPLES

TOPIC 2 — PROBABILITY BASICS

measuring how likely something is

THE DEFINITION

$$P(\text{event}) = \text{favourable outcomes} / \text{total outcomes}$$

always between 0 (impossible) and 1 (certain)

EXAMPLE 1 — one die

total outcomes : 1, 2, 3, 4, 5, 6 $\rightarrow$ 6

$$P(\text{rolling a 4}) = 1 / 6 \approx 0.167$$

$$P(\text{an even number}) = 3 / 6 = 0.5$$

THE AND RULE — independent events

$$P(A \text{ and } B) = P(A) \times P(B)$$

"independent" = one does not affect the other

example : two coins, both heads

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = 0.25$$

THE OR RULE — events that cannot both happen

$$P(A \text{ or } B) = P(A) + P(B)$$

example : a die shows 2 or 5

$$\frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

If they CAN overlap, subtract the overlap :

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

THE COMPLEMENT — the shortcut

$$P(\text{not } A) = 1 - P(A)$$

example : at least one head in two coins

$$P(\text{no heads}) = \frac{1}{4} \rightarrow 1 - \frac{1}{4} = \frac{3}{4}$$

often far easier than counting every winning case

ML LINK : a classifier outputs probabilities, not labels.

They must be between 0 and 1, and sum to 1 across classes.

Why it matters for ML: a classifier does not output a label, it outputs probabilities. They must sit between 0 and 1 and sum to 1 across the classes — which is exactly what the softmax function guarantees.

KEY POINTS FROM THIS TOPIC

Probability	favourable / total, always between 0 and 1
AND (independent)	$P(A) \times P(B)$
OR (exclusive)	$P(A) + P(B)$
OR (overlapping)	$P(A) + P(B) - P(A \text{ and } B)$
Complement	$P(\text{not } A) = 1 - P(A)$

TEST — Topic 2

1. A die is rolled. What is $P(\text{a number greater than 4})$? $\rightarrow 2/6 = 1/3$
2. Two dice are rolled. What is $P(\text{both show 6})$? $\rightarrow 1/6 \times 1/6 = 1/36$
3. Three coins are flipped. What is $P(\text{at least one head})$? $\rightarrow P(\text{no heads}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 1/8 \rightarrow 1 - 1/8 = 7/8$

Conditional probability and Bayes' theorem

Updating a belief when new evidence arrives — the most important topic in this step.

THE IDEA

$P(A | B)$ means "the probability of A, *given* that B has happened". The bar means "given". New evidence changes the probability, and **Bayes' theorem** is the rule for updating it correctly.

$$P(A | B) = P(B | A) \cdot P(A) / P(B)$$

THE CLASSIC EXAMPLE

A disease affects 1% of people. A test is 95% accurate. You test positive. Most people answer "95%". The real answer is about **16%**.

THE EASIEST METHOD — IMAGINE 1000 PEOPLE

Rather than pushing symbols around, count actual people. It gives the same answer and is far harder to get wrong.

Why a 95% accurate test gives only a 16% chance of being sick



Group	Count	Test positive
Sick (1% of 1000)	10	95% → 9.5 true positives
Healthy (the other 99%)	990	5% → 49.5 false positives
Total positives	9.5 + 49.5 = 59	

So $P(\text{sick} | \text{positive}) = 9.5 / 59 \approx \mathbf{0.16}$.

WHY SO LOW

Because the healthy group is enormous. Five percent of 990 people outnumbers ninety-five percent of only 10 people. **The rarer the condition, the more the false positives dominate** — which is why a single positive result on a rare-disease screen is usually followed by a second, different test.

WORKED EXAMPLE

TOPIC 3 — CONDITIONAL PROBABILITY & BAYES

updating a belief when new evidence arrives

CONDITIONAL PROBABILITY

$P(A | B)$ = the probability of A, GIVEN that B happened

the bar | means "given"

BAYES' THEOREM

$$P(A | B) = P(B | A) \cdot P(A) / P(B)$$

THE CLASSIC EXAMPLE — a medical test

1% of people have a disease. the test is 95% accurate.

you test POSITIVE. do you have it ?

EASIEST METHOD — imagine 1000 people

SICK : 1% of 1000 = 10 people

95% test positive → 9.5 true positives

HEALTHY : the other 990 people

5% test positive anyway → 49.5 false positives

total positive tests = 9.5 + 49.5 = 59

$$P(\text{sick} | \text{positive}) = 9.5 / 59 \approx 0.16$$

only about 16% — not 95%

WHY SO LOW ?

because the healthy group is huge. 5% of 990 people
outnumbers 95% of only 10 people.

the RARER the disease, the more false positives dominate.

ML LINK : this is why accuracy is a bad metric on rare events,
why precision and recall exist, and how spam filters and
Naive Bayes classifiers actually work.

Why it matters for ML: this is why **accuracy is a misleading metric** on rare events — a model that always predicts "healthy" scores 99% on this population while catching nobody. It is the reason precision and recall exist, and it is the entire mechanism behind Naive Bayes classifiers and spam filters.

KEY POINTS FROM THIS TOPIC

Conditional	$P(A B)$ = probability of A given B
Bayes	$P(A B) = P(B A) \cdot P(A) / P(B)$
Easiest method	imagine 1000 people and count
Key insight	rare condition → false positives dominate

TEST — Topic 3

1. 2% of a population has a condition. A test catches 90% of true cases. In 1000 people, how many are sick, and how many of those test positive? → **20 sick; 18 true positives**
2. Same setup, and the test wrongly flags 10% of healthy people. How many false positives among the 980 healthy? → **98**
3. Using both answers, what is $P(\text{sick} \mid \text{positive})$? → **$18 / (18 + 98) = 18/116 \approx 0.155$, about 16%**

Counting and combinations

How many ways can it happen?

THE IDEA

Probability needs counting: how many outcomes are there in total, and how many are favourable? Three tools cover almost everything.

Tool	Formula	Use when
Multiplication rule	$a \times b$	Choices happen in stages
Factorial	$n! = n \times (n-1) \times \dots \times 1$	Arranging all n items
Permutation	$n! / (n-r)!$	Picking r of n and order matters
Combination	$n! / [r! (n-r)!]$	Picking r of n and order does not matter

The only question to ask: does the order matter? First and second place in a race — yes, permutation. Two people on a team — no, combination. Note $0! = 1$.

WORKED EXAMPLES

TOPIC 4 — COUNTING & COMBINATIONS

how many ways can it happen ?

THE MULTIPLICATION RULE

choices in stages → **MULTIPLY** them

$$3 \text{ shirts} \times 4 \text{ trousers} = 12 \text{ outfits}$$

FACTORIAL $n!$

$n! = n \times (n-1) \times \dots \times 1$ arrangements of n items

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

note : $0! = 1$

PERMUTATION — order MATTERS

$$P(n, r) = \frac{n!}{(n-r)!}$$

example : 1st and 2nd place from 5 runners

$$5! / 3! = 120 / 6 = 20$$

COMBINATION — order does NOT matter

$$C(n, r) = \frac{n!}{[r! \cdot (n-r)!]}$$

example : choose 2 people from 5 for a team

$$5! / (2! \cdot 3!) = 120 / (2 \times 6) = 10$$

half of the permutation answer — because AB and BA are the same team but different race results.

THE ONE QUESTION TO ASK : does the ORDER matter ?

yes → permutation. no → combination.

Notice the relationship: the combination answer is always the permutation answer divided by $r!$, because each unordered group has $r!$ different orderings that all count as the same choice.

Why it matters for ML: counting underlies the binomial distribution, cross-validation splits, sampling strategies, and how many possible hyperparameter combinations a grid search must try.

KEY POINTS FROM THIS TOPIC

Multiplication	choices in stages → multiply
Factorial	$n! = n \times (n-1) \times \dots \times 1$, $0! = 1$
Permutation (order matters)	$n! / (n-r)!$
Combination (order does not)	$n! / [r! (n-r)!]$

TEST — Topic 4

1. Compute $4!$ → **24**
2. From 6 people, how many ways to choose a team of 2 (order does not matter)? → **$6!/(2! \cdot 4!) = 720/(2 \cdot 24) = 15$**
3. From 6 runners, how many ways to award 1st and 2nd place (order matters)? → **$6!/4! = 720/24 = 30$**

Random variables, distributions and expectation

From single events to whole patterns.

THE IDEA

A **random variable** is a number whose value depends on chance — a die roll, a patient's blood pressure, a model's loss on the next batch.

Type	Meaning	Examples
Discrete	Countable values	Die roll, number of patients, clicks
Continuous	Measured values	Height, temperature, loss

EXPECTED VALUE

The long-run average of a random variable — each value weighted by its probability.

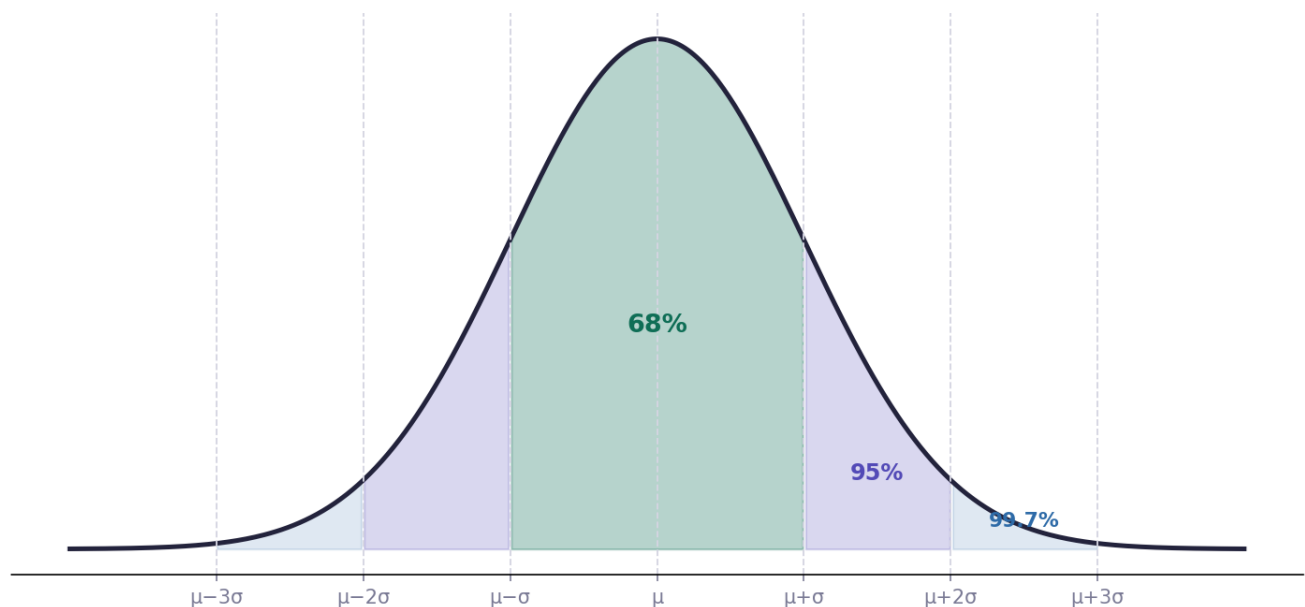
$$E[X] = \sum (\text{value} \times \text{its probability})$$

A fair die has $E[X] = 3.5$, which you can never actually roll. That is the point: the expected value is an average, not an outcome.

THE NORMAL DISTRIBUTION

The bell curve, described completely by two numbers: the mean μ (where the centre sits) and the standard deviation σ (how wide it spreads).

The normal distribution — the 68 · 95 · 99.7 rule



Within	Contains
$\mu \pm 1\sigma$	about 68% of the data
$\mu \pm 2\sigma$	about 95%

$\mu \pm 3\sigma$

about **99.7%**

WORKED EXAMPLES

TOPIC 5 — RANDOM VARIABLES & DISTRIBUTIONS

from single events to whole patterns

RANDOM VARIABLE

a number whose value depends on chance.

DISCRETE — countable (a die roll, number of patients)

CONTINUOUS — measured (height, temperature, loss)

EXPECTED VALUE $E[X]$ — the long-run average

$$E[X] = \sum (\text{value} \times \text{its probability})$$

example : a fair die

each face has probability $1/6$:

$$(1+2+3+4+5+6) \times 1/6 = 21/6 = 3.5$$

you can never roll 3.5 — it is the AVERAGE, not an outcome

A SECOND EXAMPLE — a biased coin game

win 10 with probability 0.3, lose 5 with probability 0.7

$$E[X] = (10 \times 0.3) + (-5 \times 0.7)$$

$$= 3 - 3.5 = -0.5$$

negative → on average you lose 50 cents per game

THE NORMAL DISTRIBUTION

the bell curve. described by just two numbers :

mean μ — where the centre sits

std σ — how wide the spread is

the 68 - 95 - 99.7 rule :

68% of data within 1σ of the mean

95% within 2σ 99.7% within 3σ

ML LINK : "expected loss" is literally $E[X]$. Weights are initialised from a normal distribution, and the 2σ rule is how outliers get detected.

Why it matters for ML: "expected loss" is literally $E[X]$ — the loss averaged over the data distribution. Neural network weights are initialised by drawing from a normal distribution, and the 2σ rule is a standard way to flag outliers.

KEY POINTS FROM THIS TOPIC

Random variable	a number decided by chance
Expected value	$E[X] = \sum (\text{value} \times \text{probability})$
Normal distribution	described by mean μ and std σ
68 · 95 · 99.7	within 1σ , 2σ , 3σ of the mean

TEST — Topic 5

1. A coin pays \$6 for heads and \$0 for tails. What is $E[X]$? $\rightarrow (6 \times 0.5) + (0 \times 0.5) = 3$
2. A game wins \$20 with probability 0.2 and loses \$4 with probability 0.8. Find $E[X]$ and say whether it is worth playing. $\rightarrow (20 \times 0.2) + (-4 \times 0.8) = 4 - 3.2 = 0.8$; **positive, so yes**
3. Heights are normal with $\mu = 170$ cm and $\sigma = 8$ cm. Between which two heights do about 95% of people fall? $\rightarrow \mu \pm 2\sigma = 170 \pm 16 \rightarrow 154 \text{ cm to } 186 \text{ cm}$

Master reference sheet

Topic	Item	Rule
T1	Mean	add all values, divide by how many
T1	Median	the middle value once sorted
T1	Variance	average of the squared distances from the mean
T1	Standard deviation	$\sqrt{\text{variance}}$ — back in original units
T1	Normalisation	$(x - \text{mean}) / \text{std}$
T2	Probability	favourable / total, always between 0 and 1
T2	AND (independent)	$P(A) \times P(B)$
T2	OR (exclusive)	$P(A) + P(B)$
T2	OR (overlapping)	$P(A) + P(B) - P(A \text{ and } B)$
T2	Complement	$P(\text{not } A) = 1 - P(A)$
T3	Conditional	$P(A B) = \text{probability of } A \text{ given } B$
T3	Bayes	$P(A B) = P(B A) \cdot P(A) / P(B)$
T3	Easiest method	imagine 1000 people and count
T3	Key insight	rare condition $\rightarrow$ false positives dominate
T4	Multiplication	choices in stages $\rightarrow$ multiply
T4	Factorial	$n! = n \times (n-1) \times \dots \times 1, 0! = 1$
T4	Permutation (order matters)	$n! / (n-r)!$
T4	Combination (order does not)	$n! / [r! (n-r)!]$
T5	Random variable	a number decided by chance
T5	Expected value	$E[X] = \sum (\text{value} \times \text{probability})$
T5	Normal distribution	described by mean μ and std σ
T5	68 · 95 · 99.7	within $1\sigma, 2\sigma, 3\sigma$ of the mean

VERIFY HABITS

- **Probability:** every answer must land between 0 and 1
- **Variance:** it can never be negative — squares are always positive
- **Bayes:** count people out of 1000 instead of pushing symbols
- **Counting:** ask "does order matter?" before choosing the formula
- **Combination vs permutation:** the combination answer is always smaller
- **Expected value:** the probabilities used must sum to 1

PERSONAL ERROR LOG — CARRIED FORWARD

Error	Fix
Misreading given numbers	Copy all values onto paper, labelled, before computing
Dropping a minus sign	Write subtraction as adding a negative
Assuming a shortcut applies	Check the condition first
Rushing the easy question	Same care on all three questions